

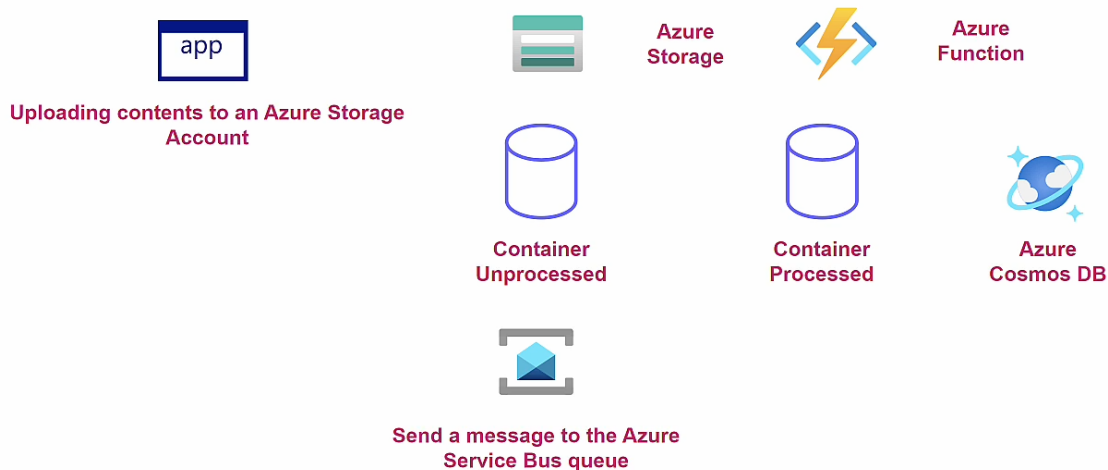


Azure Service Bus – Send and Receive Data

The process involves setting up an **Azure Service Bus** queue to manage messages about data stored in an **Azure Storage Account**. First, an application uploads files to an **"unprocessed"** container and sends corresponding messages to the **Service Bus queue**. An **Azure Function** with a **Service Bus trigger** listens for messages, retrieves the files, moves them to a **"processed"** container, and logs details in **Azure Cosmos DB**.

The end goal is to create an **automated data pipeline**, efficiently managing file processing and message-driven workflows within **Azure's cloud environment**.

1. In the last chapter, we covered Azure Service Bus, which is a messaging system on the Azure platform. To demonstrate its usage, we'll walk through a scenario where an application sends data to a container in an Azure storage account called "unprocessed" while also sending a message to an Azure Service Bus queue containing details about the uploaded object.
2. An Azure function will then read these messages, retrieve the corresponding objects from the "unprocessed" container, move them to another container called "processed," and store the object details in Azure Cosmos DB.
3. Essentially, the application sends objects to be processed into a container and messages about them to the Service Bus, while the function reads these messages, processes the objects, moves them to the "processed" container, and updates Cosmos DB with their details.



4. From our previous labs we have a storage account in place where we are going to create two containers.

+ Container
Change access level
Restore containers
Refresh
Delete
Give feedback

Search containers by prefix
Show deleted containers

Name	Last modified	Anonymous access level	Lease state	
☐ \$logs	3/12/2025, 10:45:52 AM	Private	Available	...
☐ eventhub	3/12/2025, 10:47:27 AM	Private	Available	...
☐ processed	3/12/2025, 4:55:26 PM	Private	Available	...
☐ unprocessed	3/12/2025, 4:55:20 PM	Private	Available	...

5. Now in our Azure Cosmos DB we have a container but we will create a new container.

newcosmosdb121 | Data Explorer
☆ ...

Azure Cosmos DB account

Enable Azure Synapse Link

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Quick start
- Data Explorer**
- Mirroring in Fabric (Preview)

+ New Container

Home
logdb
weblogs

6. For our new container we will use our existing database and give a name to our container then give the same partition as you can see below.

New Container



* Database id ⓘ

☐ Create new ☒ Use existing

logdb

* Container id ⓘ

bloblogs

* Partition key ⓘ

/BlobName

Add hierarchical partition key

Unique keys ⓘ

- Now you have to download two zip files attached to this lab and then extract both of them. Open the Send Messages file in Visual Studio 2022.
- Also, download these files as well we will send them to the Service bus queue.

Name	Date modified	Type	Size
 01	12-03-2025 17:12	SQL Source File	1 KB
 02	12-03-2025 17:12	SQL Source File	1 KB
 03	12-03-2025 17:12	SQL Source File	1 KB
 04	12-03-2025 17:12	SQL Source File	1 KB
 05	12-03-2025 17:12	SQL Source File	1 KB

- Now open the program here you need to give the connection string for your storage account and service bus. Then you need to specify the location of your files give the queue name and what is the container name.

```

1  using Azure.Messaging.ServiceBus;
2  using Azure.Storage.Blobs;
3  using System.IO;
4  using TransferVideos;
5
6  string blob_connection_string = "";
7  string servicebus_connection_string = "";
8
9  string container_name = "unprocessed";
10 string strPath = "C:\\tmp3";
11 string queue_name = "filequeue";
12
13 BlobServiceClient _client = new BlobServiceClient(blob_connection_string);
14 BlobContainerClient _container_client = _client.GetBlobContainerClient(container_name);
15

```

10. First open the service bus then create a new queue in it with the default settings. Open it.

 **Newbus121 | Queues** ☆ ...
Service Bus Namespace

Search Queue Refresh Give feedback

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Settings
Entities

Queues
Topics

Search to filter items by name...

Name	Status	Message count
filequeue	Active	0

11. Go to shared access policies and click on add. Give a policy name and choose to create a send policy.

 **filequeue (Newbus121/filequeue) | Shared access policies** ☆ ...
Service Bus Queue

Search Add

Overview
Access control (IAM)
Diagnose and solve problems
Service Bus Explorer
Settings

Shared access policies ☆
Properties

Search to filter items...

Policy

no policies have been set up yet.

Add SAS Policy



Service Bus

Policy name *

Sendpolicy



☐ Manage

☒ Send

☐ Listen

12. Then click on the Send policy copy one of the connection strings from here and paste it into the program.

SAS Policy: Sendpolicy



Service Bus



Regenerate primary key



Regenerate secondary key



Delete

Service Bus access control with Shared Access Signatures [Learn more](#)

Claims

☐

Manage

☒

Send

☐

Listen

Primary key

.....  

Secondary key

.....  

Primary connection string

.....  

Secondary connection string

.....  

SAS policy ARM ID

/subscriptions/d6549a66-c45c-4979-840c-3b356da446b0/resourceGroups/NewRG/... 

☐

Show AMQP connection strings

13. Now open your storage account and copy one of the connection strings from the Access keys and paste into the program.

thestorageaccount1201 | Access keys ☆ ...
Storage account

Search

- Overview
- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Data storage
- Security + networking
 - Networking
 - Front Door and CDN
 - Access keys**
 - Shared access signature
 - Encryption

Set rotation reminder Refresh Give feedback

Access keys authenticate your applications' requests to this storage account. Keep your keys in a secure location like Azure Key Vault, and replace them often with new keys. The two keys allow you to replace one while still using the other.

Remember to update the keys with any Azure resources and apps that use this storage account.
[Learn more about managing storage account access keys](#)

Storage account name
thestorageaccount1201

key1 Rotate key
Last rotated: 3/12/2025 (0 days ago)
Key
..... Show
Connection string
DefaultEndpointsProtocol=https;AccountName=thestorageaccount1201;Account... Hide

key2 Rotate key
Last rotated: 3/12/2025 (0 days ago)
Key
..... Show
Connection string
..... Show

14. Then in the program on line number 30 make sure you are changing the storage account's name.

```

24
25 BlobClient _blob_client = _container_client.GetBlobClient(Path.GetFileName(filename));
26 await _blob_client.UploadAsync(filename);
27
28 Item item = new Item();
29 item.ItemName = Path.GetFileName(filename);
30 item.ItemLocation = "https://thestorageaccount1201.blob.core.windows.net/unprocessed/" + Path.GetFileName(filename);
31
32 ServiceBusMessage _message = new ServiceBusMessage(item.ToString());
33 _message.ContentType = "application/json";
34 await _sender.SendMessageAsync(_message);
35
36 Console.WriteLine($"Uploaded blob {filename}");
37

```

15. Our every requirement is completed now we can save everything and run the program.

16. You can see that the blobs are uploaded to the container and the messages have been sent to the service bus.

```

Microsoft Visual Studio Debu x + -
Uploaded blob C:\tmp3\01.sql
Uploaded blob C:\tmp3\02.sql
Uploaded blob C:\tmp3\03.sql
Uploaded blob C:\tmp3\04.sql
Uploaded blob C:\tmp3\05.sql

D:\Azure\TransferVideos\TransferVideos\TransferVideos\bin\Debug\net6.0\TransferVideos.exe (process 31080) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

17. Now open the storage account and go to the unprocessed container there you will see your files.

unprocessed Container

Search Upload Change access level Refresh Delete Change tier Acquire lease Break lease View snapshots Create snapshot Give feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: unprocessed

Search blobs by prefix (case-sensitive) Show deleted blobs

Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
01.sql	3/12/2025, 5:25:58 PM	Hot (Inferred)		Block blob	41 B	Available
02.sql	3/12/2025, 5:26:00 PM	Hot (Inferred)		Block blob	41 B	Available
03.sql	3/12/2025, 5:26:01 PM	Hot (Inferred)		Block blob	41 B	Available
04.sql	3/12/2025, 5:26:01 PM	Hot (Inferred)		Block blob	41 B	Available
05.sql	3/12/2025, 5:26:01 PM	Hot (Inferred)		Block blob	41 B	Available

18. Open the service bus and go to the queues. Navigate to service bus explorer and here you can see your messages.

filequeue (Newbus121/filequeue) | Service Bus Explorer

Service Bus Queue

Search

Peek Mode Send messages Refresh Export messages Show message body Settings Learn more Give feedback

Queue (5) Dead-letter (0)

Peek from start Peek next messages Peek with options Re-send selected messages Download selected message body

Showing 5 of 5 messages

Sequence Number	Message ID	Enqueued Time	Delivery Count	State	Body ...	Label/Subject	Message Text
1	373dbbee037040debc9d236214...	Wed, Mar 12, 25, 05:26:00 PM G...	0	Active	109 B		["ItemName":"01.sql","ItemLocati...
2	745c2a32b5a345bebfaefc75a5f8...	Wed, Mar 12, 25, 05:26:01 PM G...	0	Active	109 B		["ItemName":"02.sql","ItemLocati...
3	26452e75ab3448f288e8012331a...	Wed, Mar 12, 25, 05:26:01 PM G...	0	Active	109 B		["ItemName":"03.sql","ItemLocati...
4	7079face984248e2b97fdccf19ad...	Wed, Mar 12, 25, 05:26:01 PM G...	0	Active	109 B		["ItemName":"04.sql","ItemLocati...
5	8c925d1d64a24c1c9ea47799c41...	Wed, Mar 12, 25, 05:26:02 PM G...	0	Active	109 B		["ItemName":"05.sql","ItemLocati...

Message Body Message Properties

Fit message body Off

["ItemName":"01.sql","ItemLocation":"https://thestorageaccount1201.blob.core.windows.net/unprocessed/01.sql"]

19. Our step 1 for the lab is completed we have successfully sent the data. Now we will move to the second part of the lab which is to receive the data.

20. Now you need to open the second program which says receive data in Visual Studio 2022. This program is based on an Azure function and it is based on a Service bus trigger. If you remember from the Azure function lab you know it works with different triggers and one of them is the service bus. In the previous lab for Azure Function, we used the event hub as a trigger.

21. This program will be activated if it sees messages in the service bus queue. It will also delete the objects from the unprocessed container and upload them to the processed container.

22. First on line number 21 we will change the blob connection string like we did before for the previous program.

```

0 references
19 public class ProcessQueueMessages
20 {
21     private static string blob_connection_string = "";
22     private static string source_container_name = "unprocessed";
23     private static string destination_container_name = "processed";
24

```

23. Then we need to change the connection string for Cosmos DB on line number 25 along with the database name and container name.

```

24
25     private static readonly string _connection_string = "AccountEndpoint=https://newcosmosdb121.documents.azure.com:443/A
26     private static readonly string _database_name = "logdb";
27     private static readonly string _container_name = "bloblogs";

```


24. Now change the name of the storage account on line number 53.

```
50
51      BlobDetails _blobdetails = new BlobDetails();
52      _blobdetails.BlobName = _message.ItemName;
53      _blobdetails.BlobLocation = "https://thestorageaccount1201.blob.core.windows.net/processed/"
54      _blobdetails.ContentLength = _info.ContentLength.ToString();
55      _blobdetails.LastModified = _info.Details.LastModified.ToString();
56      _blobdetails.id = Guid.NewGuid().ToString();
```

25. In the end go to the local.settings.json file to change the connection string of your queue.

```
BlobDetails.cs  Item.cs  ProcessFiles.csproj  local.settings.json*  ProcessQueueMessages.cs*
Schema: https://json.schemastore.org/local.settings.json
1  {
2    "IsEncrypted": false,
3    "Values": {
4      "AzureWebJobsStorage": "UseDevelopmentStorage=true",
5      "FUNCTIONS_WORKER_RUNTIME": "dotnet",
6      "queueconnection": ""
7    }
8  }
```

26. To get the connection string from the queue this time you need to add a new policy for Listening and then copy one of the connection string .

The screenshot shows the Azure portal interface. On the left, the 'filequeue (Newbus121/filequeue)' queue is selected under 'Queues > filequeue (Newbus121/filequeue)'. The 'Shared access policies' section is visible, showing a table with two policies: 'SendPolicy' and 'ListenPolicy'. The 'ListenPolicy' policy is selected, and its configuration is shown on the right. The 'SAS Policy: ListenPolicy' configuration includes options to regenerate keys, delete the policy, and a table of claims. The 'Listen' claim is checked. Below the claims, there are fields for the primary and secondary keys, primary and secondary connection strings, and the SAS policy ARM ID.

27. Remember to delete the entity path from the connection string.

```
1  {
2    "IsEncrypted": false,
3    "Values": {
4      "AzureWebJobsStorage": "UseDevelopmentStorage=true",
5      "FUNCTIONS_WORKER_RUNTIME": "dotnet",
6      "queueconnection": "Endpoint=sb://newbus121.servicebus.windows.net/;SharedAccessKeyName=ListenPolicy;SharedAccessKey=AV3m5Vfq"
7    }
8  }
```

28. Once you have done everything just save all and run the program. Here we can see that the data has been created.

```
C:\Users\PULKIT\AppData\Local\Microsoft\Windows\CurrentVersion\Explorer\Recent\
[2025-03-12T12:17:53.407Z] Executing 'ProcessMessages' (Reason='(null)', Id=a8e0621b-59d1-43a6-b689-d7925d27731c)
[2025-03-12T12:17:53.413Z] Trigger Details: MessageId: 7079face984248e2b97fdccf19add6f4, SequenceNumber: 4, DeliveryCount: 1, EnqueuedTimeUtc: 2025-03-12T11:56:01.7780000+00:00, LockedUntilUtc: 2025-03-12T12:18:52.9330000+00:00, SessionId: (null)
[2025-03-12T12:17:55.559Z] 12-03-2025 11:56:00 +00:00
[2025-03-12T12:17:55.559Z] 12-03-2025 11:55:58 +00:00
[2025-03-12T12:17:55.559Z] 12-03-2025 11:56:01 +00:00
[2025-03-12T12:17:55.559Z] 12-03-2025 11:56:01 +00:00
[2025-03-12T12:17:55.560Z] 41
[2025-03-12T12:17:55.561Z] 41
[2025-03-12T12:17:55.562Z] 41
[2025-03-12T12:17:55.560Z] 41
[2025-03-12T12:17:55.559Z] 12-03-2025 11:56:01 +00:00
[2025-03-12T12:17:55.564Z] 41
Item created
Item created
Item created
Item created
Item created
[2025-03-12T12:17:58.642Z] Executed 'ProcessMessages' (Succeeded, Id=28187d46-116d-475a-9061-30b2cb3fe47e, Duration=5279 ms)
[2025-03-12T12:17:58.642Z] Executed 'ProcessMessages' (Succeeded, Id=c12cd63b-1a6b-4923-ac7a-0d1703e94696, Duration=5279 ms)
[2025-03-12T12:17:58.642Z] Executed 'ProcessMessages' (Succeeded, Id=a8e0621b-59d1-43a6-b689-d7925d27731c, Duration=5279 ms)
[2025-03-12T12:17:58.642Z] Executed 'ProcessMessages' (Succeeded, Id=9cae6518-7291-4127-8649-995f73be62f8, Duration=5279 ms)
[2025-03-12T12:17:58.642Z] Executed 'ProcessMessages' (Succeeded, Id=8a0cde38-aeb5-41e4-93fb-a4ec84613bfb, Duration=5279 ms)
```

29. Let's open the processed container first, we have the files here. We can also see that the files have been deleted from the unprocessed container.

processed

Container

Search

Upload Change access level Refresh Delete Change tier Acquire lease Break lease View snapshots Create snapshot Give feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: processed

Search blobs by prefix (case-sensitive)

Show deleted blobs

Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
01.sql	3/12/2025, 5:47:55 PM	Hot (Inferred)		Block blob	41 B	Available ***
02.sql	3/12/2025, 5:47:55 PM	Hot (Inferred)		Block blob	41 B	Available ***
03.sql	3/12/2025, 5:47:55 PM	Hot (Inferred)		Block blob	41 B	Available ***
04.sql	3/12/2025, 5:47:55 PM	Hot (Inferred)		Block blob	41 B	Available ***
05.sql	3/12/2025, 5:47:55 PM	Hot (Inferred)		Block blob	41 B	Available ***

unprocessed

Container

Search

Upload Change access level Refresh Delete Change tier Acquire lease Break lease View snapshots Create snapshot Give feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: unprocessed

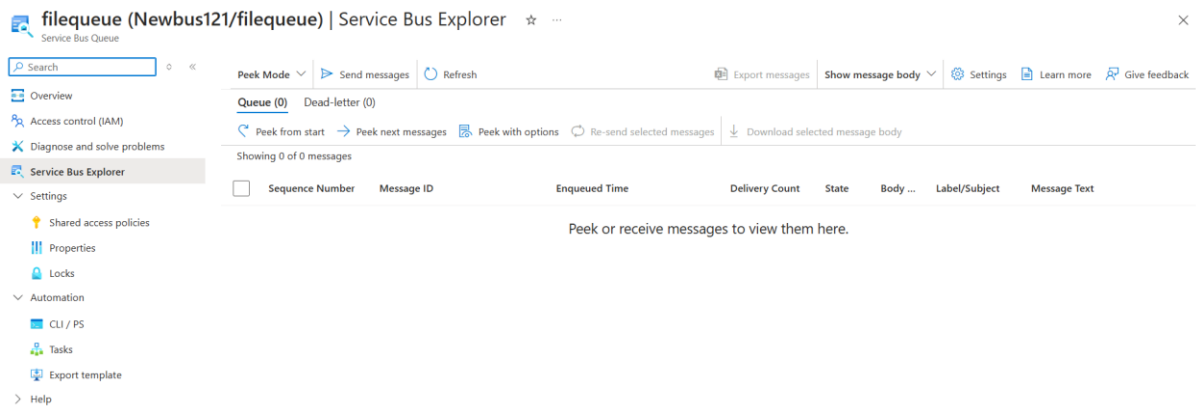
Search blobs by prefix (case-sensitive)

Show deleted blobs

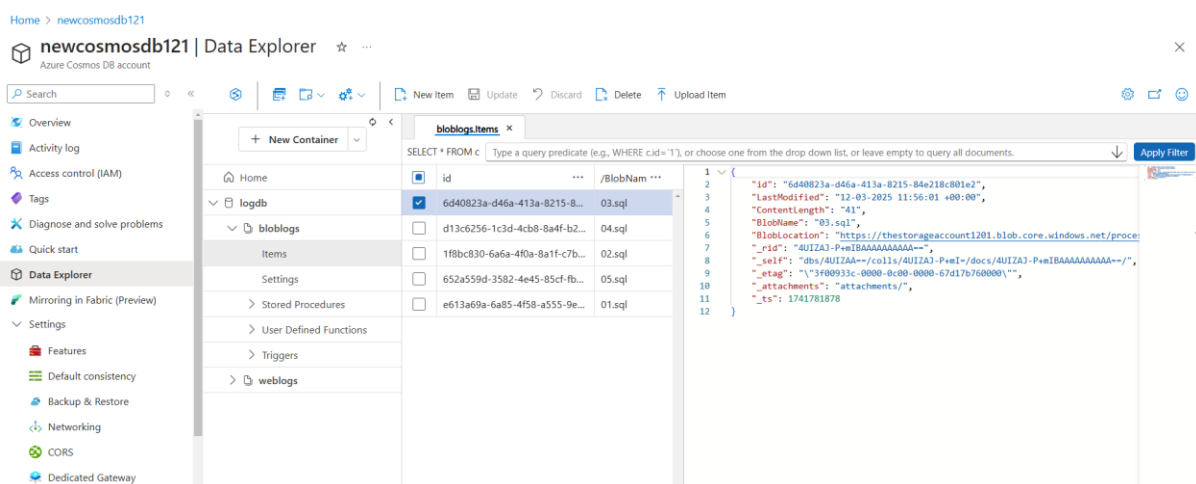
Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
No results						

30. In the service bus queue we can see that the messages have been processed.



31. Let's open the data explorer in the Cosmos DB. In our container here we can see that the items have been added and we can see their information.



32. Once you are done with the lab just delete all of the resources.